4.42*10^113Pa

Appendix I

Proof of Quantum Value of Young's Modulus of Space Space-Time Obtained in Tables 16.1 and 16.2

I.1 Introduction

The purpose of this appendix is to verify the results and the formulas obtained by iterations in tables 16.1 and 16.2 of chapter 16 of this book from the value of Young's modulus given in the publication of T.G. Tenev and M.F. Horstemeyer The Mechanics of Spacetime. A Solid Mechanics Perspective on the Theory of General Relativity chapter 4 Summary and Conclusion.

In this publication, the quantum Young's modulus of space-time takes the following expression:

$$Y = \frac{6c^7}{2\pi\hbar G^2}$$

Close

The numerical application gives:

$$Y = \frac{6(299\,792\,458)^7}{2\pi\left(\frac{6.62607015\times10^{-34}}{2\pi}\right)\times(6.6742\times10^{-11})^2} = 4.42\times10^{113}\,\text{Pa}$$

I.2 Determination of Young's Modulus Formula from the Expressions of the Table 16.1 of Chapter 16

We have demonstrated in chapter 16 the following expression:

$$\rho\, Vc^2 = \frac{Gh^2}{4\pi\, Vc^2}$$